

From Dome to the Mall: What Accra's Food Markets Reveal About Trade-off, Tech, & Tomorrow

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Research Question 1

What truly shapes food access in Accra beyond simple market connectivity?

Background

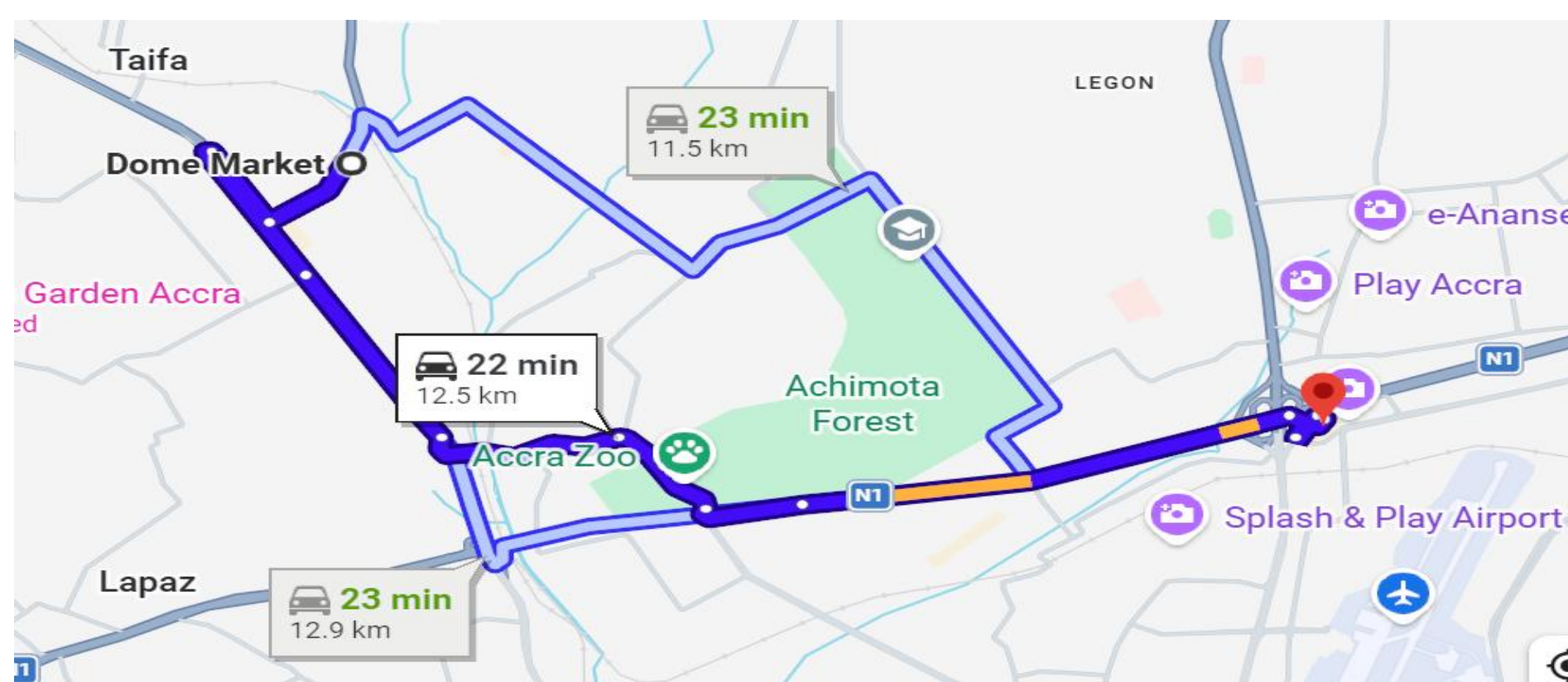
Food access in Accra is not a question of market connectivity, it is shaped by trade-offs between affordability, flexibility, and convenience. The city operates two food systems simultaneously: vibrant informal open markets and expanding modern retail, each serving different economic realities. Ghana's informal sector employs over **80% of the national workforce**, dominating food retail and supplying most urban households. (*Osei-Boateng & Ampratwum, 2011*) Despite their central role, informal vendors in Accra face evictions, and neglect of market infrastructure, paying fees while receiving little in return. (*Anyidoho & Steel, 2015*)

Main Argument:

Food access in Accra is shaped by trade-offs between affordability, flexibility, and convenience across informal and formal systems where Agri-tech offers potential solutions.

Methodology

- **Interviews:** 4 in-depth interviews with sellers and consumers at Dome Market and Accra Mall
- **Observations:** Field Observation at Dome Market (informal) and Accra Mall (modern retail), ~ 7.8 miles apart
- **Desk research:** Literature on Ghana's informal economy, cold chain investment, and agricultural technology



A 22-minute drive that crosses an economic divide

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Research Question 2

How do informal open markets and modern retail differ in pricing, supply, and livelihoods?

The Informal Market: Dome Market



A picture of me at Dome Market, Accra, Ghana

SUPPLY

Strong local networks, no major access issues. Known farm-to-market routes

PRICING

Flexible and negotiated daily, Seasons dictate prices — rainy cheaper, dry expensive

PERISHABILITY

No cold storage, Tomatoes spoil in 3–4 days. Losses are absorbed or thrown away

LIVELIHOODS

Survival-oriented income, Sellers aim to recover cost, Credit-based buying is common

Dome Market is one of Accra's most active informal trading hubs, located in the Ga East Municipal District. It is a full-sensory, open-air marketplace where hundreds of vendors, most of them women who sell fresh produce, spices, grains, and household goods daily.

Produce is sourced from established local supply routes: onions from Adjen Kotoku, tomatoes from Agboghloshie. Prices are negotiated, quantities are flexible, and transactions are built on trust and long-standing relationships between buyers and sellers.



Field interview with Patricia Didongo and Mary Ayesi

KEY INSIGHT

Market access is not the problem. The core issue is perishability and price volatility, both driven by the total absence of cold storage. Losses are silently absorbed or thrown away.

"Sometimes the market is good, sometimes very bad."— Mary Ayesi, seller, Dome Market

"During rainy season, more quantity and cheaper. During dry season, very expensive."— Patricia Didongo, seller, Dome Market

The Formal System: Accra Mall

Modern retail operates on a different logic — warehouse-backed supply, daily restocking from international sources, and fixed pricing. These features ensure consistency and year-round availability but create a new problem: price exclusion.

SUPPLY CHAIN

Warehouse + daily deliveries, Heavy import reliance products sourced from South Africa

STORAGE

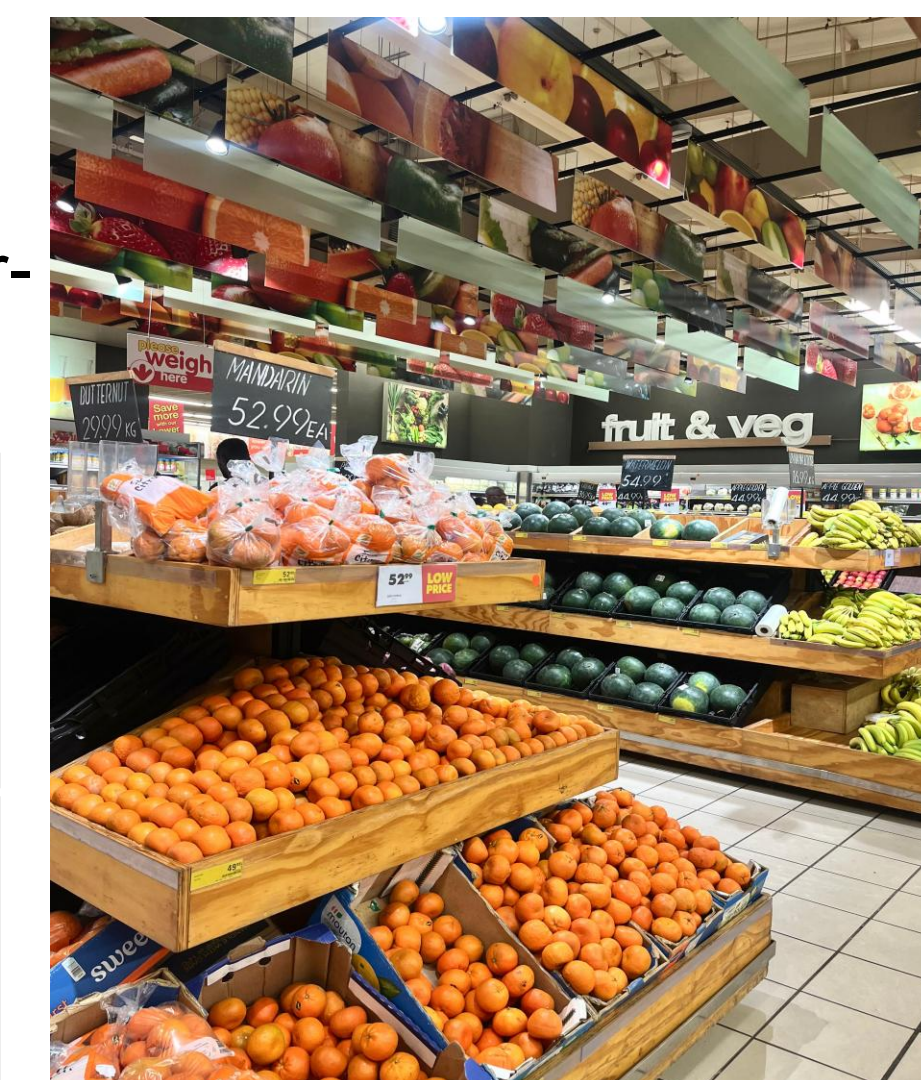
Cold chain infrastructure eliminates spoilage — a structural advantage informal markets lack

PRICING

Fixed, non-negotiable, Premium prices limit access for most of the Accra's population

VARIETY

Broader range, available off-season, Shortages occur when international supply lines break



"Most of our products come from South Africa... even out of season, you can still get it."— Saviour, grocery store manager, Accra Mall

KEY INSIGHT

Modern retail solves perishability but creates exclusion. When formal retail imports what local farmers grow, it disconnects from the local food economy entirely.

Consumer Perspective

"Everything is about pricing. People don't really look at brands. Markets give more control."— Theophilus K.K. Sarpong, consumer, Accra Mall

KEY INSIGHT

Price brings people to the market. Convenience brings them to the mall.

Research Question 3

Can Agri-tech meaningfully close the gaps in Accra's foods system?

The Tomato Crisis: A Case Study in Post-Harvest Loss

Ghana produces ~420,000 MT of tomatoes annually yet loses approximately **40% to post-harvest spoilage** which is driven by poor handling, absence of cold storage, and weak haulage infrastructure. In Agboghloshie alone, over 12,600 MT of tomatoes valued at more than \$31 million is lost annually. (*ASNAPP, 2020*)

Ghana tomato: total production vs. available to market



"Tomatoes start rotting after a few days, we just throw it away."— Mary Ayesi, Dome Market

Can Agri-Tech Bridge the Gap?



According to Agri Impact Ghana, Ghana's food system challenge is not a production problem, it is a **system design and coordination problem**.

"Ghana does not primarily have a production problem — it has a system design and coordination problem."— Lary Selorm Agri Impact Ghana

The most impactful near-term intervention: **low-cost solar-powered cooling at the market level**. A single 40-foot cold storage unit at Ashaiman Market projects over \$73,000 net profit in year one. (*ASNAPP, 2020*)

- **Solar Cold Storage** — Zero-energy cool chambers and solar cold rooms for community market centers
- **Digital Platforms** — Mobile apps connecting farmers directly to vendors, reducing gluts and stabilizing prices
- **Trader Support** — Moving beyond farm-level interventions to equip informal market vendors with tools, training, and resources to reduce losses and improve their income stability

Conclusion

Informal vendors and modern retail are both essential to how Accra eats but spoilage, price instability, and absent cold storage silently erode vendor incomes every day. Agri-tech solutions are already emerging in Greater Accra, but lasting change requires investment, policy support, and informal vendors at the center of the solution.